

Northern University of Business and Technology Khulna
Department of Computer Science and Engineering

[Project Proposal]

Project Title:
AI Campus Assistant

Submitted by:
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1. Project Title:

AI Campus Assistant

2. Problem Statement

In many universities, students frequently encounter challenges such as:

- Difficulty finding class and examination schedules
- Missing assignment deadlines
- Lack of centralized academic information
- Limited access to personalized academic support
- Difficulty understanding complex academic topics
- Inefficient communication between students and teachers

Traditional academic management systems mainly provide static information and often lack intelligent assistance. Students must search through multiple sources to find information and receive academic support.

The proposed AI Campus Assistant aims to solve these issues by integrating campus management features with artificial intelligence to create a centralized and intelligent academic support system.

3. Objectives

The primary objectives of this project are:

General Objectives

- Develop an AI-powered academic assistance platform
- Centralize academic information management
- Improve communication between students, teachers, and administrators
- Enhance student learning experiences through AI technologies

Specific Objectives

- Provide class and examination routine management
- Manage assignments and CT schedules
- Enable role-based access control
- Integrate AI-powered chatbot functionality
- Generate personalized study plans
- Summarize lecture notes automatically
- Generate quizzes using artificial intelligence
- Deploy the system online for public accessibility

4. Proposed Features

4.1 Authentication System

Features:

- Student Registration
- Student Login
- Teacher Login
- Admin Login
- Password Encryption
- Session Management
- Role-Based Access Control

4.2 Academic Information Management

Features:

- Class Routine Management
- Examination Routine Management
- Course Management
- Academic Notice Management

4.3 Assignment Management

Features:

- Assignment Creation
- Assignment Tracking
- CT Schedule Management
- Assignment Deadlines
- Assignment Status Monitoring

4.4 Student Dashboard

Features:

- Upcoming Classes
- Upcoming Exams
- Pending Assignments
- Academic Statistics
- Personalized Notifications

4.5 AI Chat Assistant

Students can interact with an AI chatbot to:

- Ask academic questions
- Receive learning guidance
- Understand difficult concepts
- Obtain academic information

Example: ‘What is Object-Oriented Programming?’

The AI will provide an educational explanation.

4.6 Rule-Based Academic Query System

The system will answer institutional queries using database information.

Examples:

- What is the routine of Section 7C?
- When is the next DBMS examination?
- What assignments are due this week?

The system will retrieve information directly from the database and present it in a structured format.

4.7 AI Study Planner

Students can generate personalized study plans.

Example:

Input: ‘Create a study plan for the Data Structures exam within 7 days.’

Output:

- Day-by-day study schedule
- Recommended topics
- Suggested revision timeline

4.8 AI Quiz Generator

Students and teachers can generate quizzes from topics.

Example:

Input:

Artificial Intelligence

Output:

- Multiple Choice Questions
- True/False Questions
- Short Questions

4.9 AI Note Summarizer

Students can upload notes or paste content.

The AI will:

- Summarize lengthy notes
- Extract important concepts
- Highlight key points

4.10 Notification System

Features:

- Assignment Notifications
- Examination Notifications
- Routine Update Notifications
- Academic Alerts

5. Technology Stack

Component	Technology
Frontend	React.js
UI Framework	Tailwind CSS
Backend	Node.js
API Framework	Express.js
Database	MySQL
AI Service	Gemini API
Authentication	JWT
Version Control	Git & GitHub
API Testing	Postman
Deployment (Frontend)	Vercel
Deployment (Backend)	Render

N.B.: The proposed technology stack may be changed or updated according to project requirements during development.

6. Expected Outcome

After successful completion of the project, the system will:

- Provide centralized academic information management
- Improve communication among stakeholders
- Reduce academic information confusion
- Increase student productivity
- Provide intelligent academic support
- Automate routine academic tasks
- Enhance learning through AI-powered assistance

The AI Campus Assistant will serve as a modern educational support platform that combines academic management and artificial intelligence to improve the overall learning experience.

